Knox Design Analysis Notebook

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EC552 - Computational Synthetic Biology for Engineers

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Imports

In [1]:
```
import requests
import csv
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import json

from sklearn import tree
from sklearn.model_selection import train_test_split
```

In [ ]:
```
response = requests.get(url + '/designSpace/listGroupSpaces?groupID=TestGroup1')
print(response.text)
```

Knox Request Functions

In [2]:
```
url = 'http://localhost:8080'

def ruleEvaluateByGroup(evalName, groupID, ruleGroupID, labelingMethod="sign"):
    """
    API Request to Knox to run Rule Evaluation Algorithm.

    Returns:
    metrics (list of pandas DataFrames): purity_metrics, designToRule
    """
    response = requests.post(url + '/rule/evaluate?' +
                             "evaluationName=" + evalName + '&' +
                             "designGroupID=" + groupID + '&' +
                             "rulesGroupID=" + ruleGroupID + '&' +
                             "labelingMethod=" + labelingMethod
    )

    return processRuleEval(response)


def ruleEvaluateByDesigns(evalName, designIDs, ruleGroupID, designScores, labeli
    designSpaceIDs = listToStringList(designIDs)
    designScoresStr = listToStringList(designScores) # designScores should be a

    # Submit Request
    response = requests.post(url + '/rule/evaluate?' +
                             "evaluationName=" + evalName + '&' +
                             "designSpaceIDs=" + designSpaceIDs + '&' +
                             "rulesGroupID=" + ruleGroupID + '&' +
                             "designScores=" + designScoresStr + '&' +
                             "labelingMethod=" + labelingMethod
    )

    return processRuleEval(response)


def getRuleEvaluation(evalName):
    response = requests.get(url + '/rule/getEvaluation?' + "evaluationName=" + e

    return processRuleEval(response)


def deleteRuleEvaluation(evalName):
    response = requests.delete(url + '/rule?' + "evaluationName=" + evalName)

    if not response.text:
        return f'"{evalName}" Sucessfully Deleted'
    else:
        return response.text


def processRuleEval(response):
    # Change to Pandas DataFrame
    json_data = json.loads(response.text)

    purity_metrics_df = pd.DataFrame(json_data["evaluationResults"]).T

    designToRule_df = pd.DataFrame(json_data["designToRule"], index=json_data["d
    cols = designToRule_df.columns.to_list()
    cols.remove("labels")
    cols.remove("scores")
    cols.remove("designIDs")
    cols = ["labels", "scores"] + cols
    designToRule_df = designToRule_df[cols]
```

```
    designToRule_df = designToRule_df[cols]

    return purity_metrics_df.sort_values("impact"), designToRule_df.sort_values(


def listToStringList(list_input):
    return ",".join(list_input)
```

Decision Tree Functions

In [3]:
```
def exampleTree(X, y, **kwargs):
    # Mess around with different parameters

    dt_clf = tree.DecisionTreeClassifier(
        splitter=kwargs.get('splitter', 'best'),
        max_depth=kwargs.get('max_depth', None),
        min_samples_split=kwargs.get('min_samples_split', 400),
        min_samples_leaf=kwargs.get('min_samples_leaf', 200),
        max_features=kwargs.get('max_features', None),
        max_leaf_nodes=kwargs.get('max_leaf_nodes', None)
    )

    dt_clf = dt_clf.fit(X, y)

    return dt_clf

## Build more Trees, multiclassification and regression
```

Design Analysis

Run RuleEvaluation, debug, check dataframes, save CSVs.

In [ ]:
```
#deleteRuleEvaluation('test')
```

In [12]:
```
response = requests.post(url + '/rule/evaluate?' +
                            "evaluationName=test2" + '&' +
                            "designGroupID=TestGroup1" + '&' +
                            "rulesGroupID=TestRules2" + '&' +
                            "labelingMethod=median"
  )
json_data = json.loads(response.text)
print(list(json_data.keys()))
print(list(json_data["evaluationResults"].keys())[:3])

# Check what columns exist in the evaluation results
first_key = list(json_data["evaluationResults"].keys())[0]
print(json_data["evaluationResults"][first_key])
```

```
['evaluationResults', 'designToRule']
['MSW_PB2_B1_Rule_I', 'MSW_PB1_Rule_R_v1', 'MSW_B1_S1_Rule_B_v1']
{'totalImperfection': False, 'ruleSpaceID': 'MSW_PB2_B1_Rule_I', 'goodDesignsEli
m': 0, 'impact': 0, 'totalPerfection': True, 'numIncorrect': 3996, 'poorDesignsE
lim': 0, 'totalDesignsElim': 0, 'poorPerfection': False, 'poornessPercent': 100,
'numCorrect': 3996, 'goodPerfection': False, 'goodnessPercent': 100, 'poorElimin
ationPercent': None}
```

In [4]:
```
purity_metrics_df, designToRule_df = getRuleEvaluation('test2')
```

In [7]:
```
purity_metrics_df.to_csv("C:/Users/black/Documents/Knox_BE552/purity_metrics.csv
```

In [9]:
```
purity_metrics_df
```

Out[9]:

	totalImperfection	ruleSpaceID	goodDesignsElim	impact	tot
MSW_P2_Rule_M_v1	False	MSW_P2_Rule_M_v1	3514	-51	
MSW_PP2_Rule_M_v1	False	MSW_PP2_Rule_M_v1	3584	-38	
MSW_PH1_Rule_NI_v1	False	MSW_PH1_Rule_NI_v1	2130	-32	
MSW_PIn1_PH1_Rule_O	False	MSW_PIn1_PH1_Rule_O	2130	-32	
MSW_P1_Rule_M_v1	False	MSW_P1_Rule_M_v1	3048	-28	
...	...	...	...	...	
MSW_PP2_Rule_NI_v1	False	MSW_PP2_Rule_NI_v1	412	38	
MSW_PIn1_PP2_Rule_O	False	MSW_PIn1_PP2_Rule_O	412	38	
MSW_P2_S1_Rule_F_v1	False	MSW_P2_S1_Rule_F_v1	482	51	
MSW_P2_P1_Rule_A_v1	False	MSW_P2_P1_Rule_A_v1	482	51	
MSW_P2_Rule_NI_v1	False	MSW_P2_Rule_NI_v1	482	51	

210 rows × 14 columns

In [8]:
```
designToRule_df.to_csv("C:/Users/black/Documents/Knox_BE552/designToRule.csv")
```

In [10]:
```
designToRule_df
```

Out[10]:

	labels	scores	MSW_PB2_B1_Rule_I	MSW_PB1_Rule_R_v1	MSW_B1_S
MSW_design_(181)_v1	0	-0.955420	0	0	
MSW_design_(181)	0	-0.955420	0	0	
MSW_design_(284)	0	-0.948819	0	0	
MSW_design_(284)_v1	0	-0.948819	0	0	
MSW_design_(277)	0	-0.944792	0	0	
...	...	...	...	...	
MSW_design_(3929)_v1	1	3.437174	0	0	
MSW_design_(3927)_v1	1	3.447912	0	0	
MSW_design_(3927)	1	3.447912	0	0	
MSW_design_(3930)	1	3.462357	0	0	
MSW_design_(3930)_v1	1	3.462357	0	0	

7992 rows × 212 columns

In [ ]:
```
# Use this for (gnn_predicted_scores.csv) designs already have the attached scor
#evalName = 'test'
#groupID = 'TestGroup1'
#ruleGroupID = 'TestRules2'
#labelingMethod = 'median'
#purity_metrics_df, designToRule_df = ruleEvaluateByGroup(evalName, groupID, rul
```

Extract Features and Labels from designToRule_df

In [11]:
```
feature_names = designToRule_df.columns.to_list()[2:]
X = designToRule_df.iloc[:, 2:].to_numpy(dtype=int)
y_labels = designToRule_df["labels"].to_numpy(dtype=int)
y_scores = designToRule_df["scores"].to_numpy(dtype=float)
```

Train - Test Split

In [14]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y_labels, test_size=0.3,
X_train_r, X_test_r, y_train_r, y_test_r = train_test_split(X, y_scores, test_si
```

In [22]:
```
X_train
```

Out[22]:
```
array([[0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 1, 1, 0],
       ...,
       [0, 0, 0, ..., 1, 0, 0],
       [0, 0, 0, ..., 1, 0, 0],
       [0, 0, 0, ..., 1, 0, 0]])
```

In [24]:
```
y_train
```

Out[24]:
```
array([1, 0, 1, ..., 0, 1, 1])
```

In [26]:
```
y_train_r
```

Out[26]:
```
array([ 1.8393863 , -0.6781499 ,  2.0374103 , ..., -0.68347406,
        2.5277047 ,  2.2762978 ])
```

Build Trees

In [29]:
```
# Binary Classification Tree
from sklearn.tree import DecisionTreeClassifier, DecisionTreeRegressor

binary_clf = DecisionTreeClassifier(max_depth=5, min_samples_split=400, min_samp
binary_clf.fit(X_train, y_train)
```

Out[29]:
```
▾ DecisionTreeClassifier
DecisionTreeClassifier(max_depth=5, min_samples_leaf=200, min_samples_s
plit=400,
                       random_state=42)
```

In [30]:
```
  # Regression Tree
reg_tree = DecisionTreeRegressor(max_depth=5, min_samples_split=400, min_samples
reg_tree.fit(X_train_r, y_train_r)
```

Out[30]:
```
▾ DecisionTreeRegressor
DecisionTreeRegressor(max_depth=5, min_samples_leaf=200, min_samples_sp
lit=400,
                      random_state=42)
```

In [31]:
```
# Multi-Classification Tree (bin scores into quartiles)
# Use integer labels to avoid sklearn alphabetical sorting issues
y_quantile = pd.qcut(y_scores, q=4, labels=[0, 1, 2, 3])  # 0=awful, 1=poor, 2=g
X_train_m, X_test_m, y_train_m, y_test_m = train_test_split(X, y_quantile, test_
```

In [32]:
```
y_test_m
```

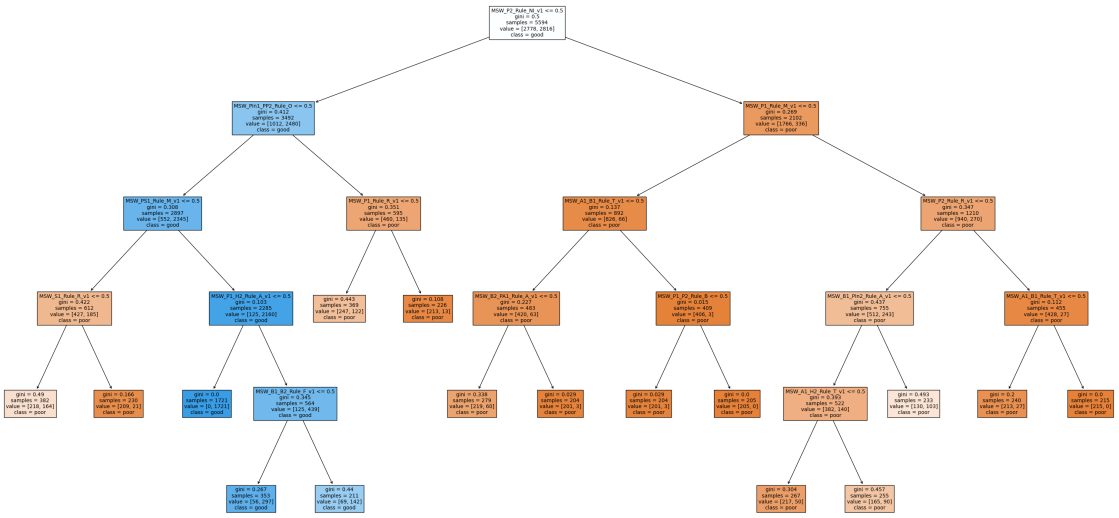
Out[32]:
```
[0, 2, 1, 2, 3, ..., 3, 1, 3, 2, 2]
Length: 2398
Categories (4, int64): [0 < 1 < 2 < 3]
```

In [33]:
```
multi_clf = DecisionTreeClassifier(max_depth=5, min_samples_split=400, min_sampl
multi_clf.fit(X_train_m, y_train_m)
```

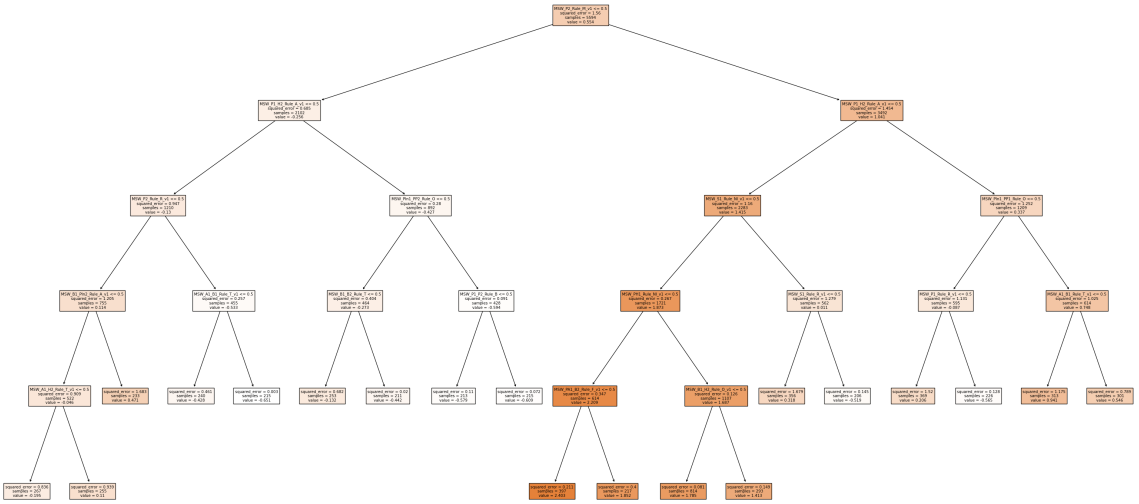
Out[33]:
```
▾ DecisionTreeClassifier
DecisionTreeClassifier(max_depth=5, min_samples_leaf=200, min_samples_s
plit=400,
                       random_state=42)
```

Plot Trees

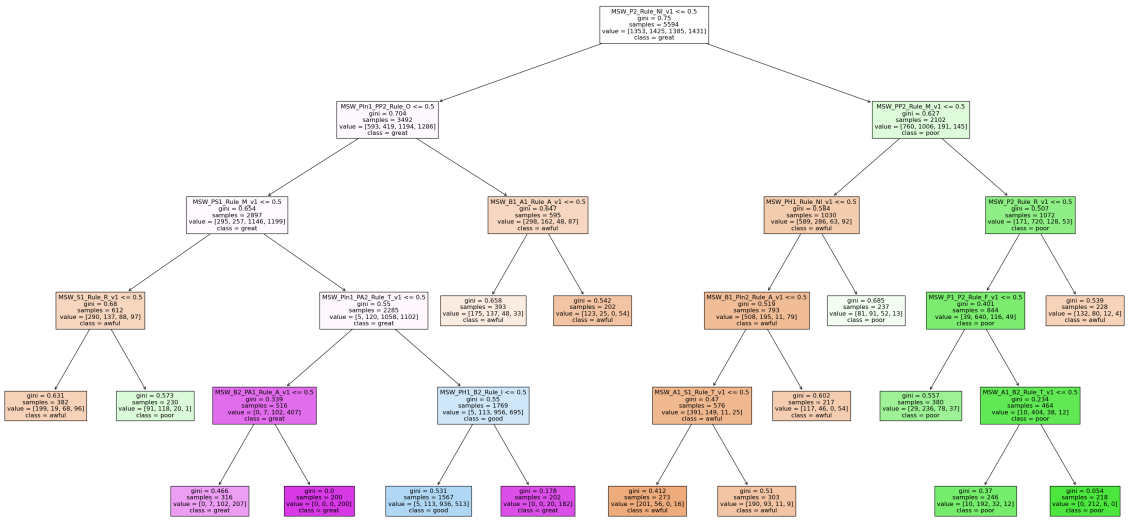
In [34]:
```
fig, ax = plt.subplots(figsize=(40, 20))
tree.plot_tree(binary_clf, feature_names=feature_names, class_names=["poor", "go
fig.savefig("binary_classification_tree.png", dpi=150, bbox_inches='tight')
```



In [35]:
```
fig, ax = plt.subplots(figsize=(40, 20))
tree.plot_tree(reg_tree, feature_names=feature_names, filled=True, ax=ax)
fig.savefig("regression_tree.png", dpi=150, bbox_inches='tight')
```



In [36]:
```
fig, ax = plt.subplots(figsize=(40, 20))
tree.plot_tree(multi_clf, feature_names=feature_names, class_names=["awful", "po
ax=ax)
fig.savefig("multi_classification_tree.png", dpi=150, bbox_inches='tight')
```



Save Trees

In [ ]:
```
# Done above
```

Save Purity Metrics and DesignToRule DF to CSV

In [ ]:
```
# Done above
```

Deliverables

1. Explanation of rule selection:

In [ ]:

Rule Type	Count	Why
R (Do Not Repeat)	19	Tests if repeating any specific part hurts/helps p parts for complete coverage
M (Must Include)	19	Tests if the presence of each part correlates with
NI (Not Include)	19	The complement of M — tests if excluding a part co
S (Starts With)	19	Tests if positional placement at the start matters
E (Ends With)	19	Tests if positional placement at the end matters f
B (Before)	42	Tests ordering constraints — all 36 CDS pairwise c cross-type pairs, since part ordering is fundament
I (Interference)	42	Tests if adjacent placement of two parts is harmfu checks direct upstream interference
F (Follows)	15	Tests if direct adjacency of specific parts is ben promoter-CDS pairs
A (After)	15	Reverse of F — tests direct downstream adjacency
T (Together)	15	Tests co-occurrence — do certain parts need to app promoter-promoter, and cross-type pairs
O (Not Orthogonal)	15	Tests mutual exclusivity — if one part is present,

```
Why these rules: We chose to cover all single-part rules exhaustively (every p
individual part effect is missed. For pairwise rules, we emphasized B and I wi
ordering and junction interference are the most biologically relevant constrai
remaining pairwise rules (F, A, T, O) use smaller representative subsets to ad
 the decision trees a diverse feature set to split on.
```

1. What are we learning from each tree type?

We are assessing the performance of rules with different granularities of circuit score as a metric. Regression tree assesses performance on the continuum of GNN data, while binary just does so in terms of good and bad, and multi-class splits good and bad further into: "awful", "poor", "good", "great".

- Binary classification — which rules most clearly separate good from poor designs.
- Regression — predicts the actual score, so you learn which rules have the biggest quantitative impact on performance, not just good/bad but how much better
- Multi-classification — reveals if certain rules only matter at the extremes

1. What are some interesting insights you noticed about some of the rules?

- Based on the binary tree, P2 not being present generally produces better designs
- The regression tree also captures the importance of P2, where P2 not being included is the most distinguishing feature from designs being scored well vs. poorly.
- This is also represented in the multi-class tree, but we don't get many interesting insights at the extremes for the poorly-performing P2-containing designs as gini scores don't change nearly as dramatically as in the good designs.
- For example: Pin1_PA2_Rule_T splits already great designs into good and great (gini: 0.55 vs gini: 0.34), where designs with PIn1 and PA2 together perform the best.

1. If you were to experimentally validate five of these rules, which would you choose? Why? Rules near the top of the tree give the most information on how to split the design space so these rules should be the first target to validate experimentally. We could choose one tree and take the rules in the top 2-3 layers to validate or give each rule a score based on its proximity to the root node across the three trees and take the top 5 from there.

Claude Code (CLI tool) was used throughout this assignment for:

- **GOLDBAR Generator CSV creation**: A Python script was generated to programmatically create all pairwise part combinations for the rule columns (B, I, F, A, T, O) and populate single-part rules (R, M, NI, S, E) for all 19 parts, rather than manually writing 200+ rule entries by hand.
- **Debugging Knox API issues**: Diagnosing 500 errors from the rule evaluation endpoint, checking Neo4j database state, and identifying mismatched group IDs and column name casing (`"Impact" vs "impact"`).
- **Sklearn boilerplate**: Generating the DecisionTreeClassifier/Regressor setup, train-test split, and plot_tree calls. Identified and fixed a label ordering bug where sklearn's alphabetical sorting of string class names caused incorrect labels in the multi-classification tree.
- **GOLDBAR language understanding**: Interpreting rule types, writing example GOLDBAR expressions, and understanding graph operations (merge, and, or, join) for the walkthrough portion.

  All tree analysis, interpretation of results, and written deliverable answers reflect our own understanding of the data and rules.

In [ ]: